

Supplementary Material

Promotional effect of Mn-doping on the structure and performance of spinel ferrite microspheres for CO hydrogenation

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Fig. S3. Iron-time-yield (FTY), product distribution and C₂=-C₄= olefin space-time-yield (STY) during a temperature-scanning experiment in the range 493-613K over a Fe-13Mn catalyst. Reaction conditions: 25mg catalyst diluted with SiC (1:3), 2.0 MPa, GHSV=4000 h⁻¹, TOS= 20h, H₂/CO=1.

Table S1Quantitative H₂ consumption area of Fe-Mn microspheres during TPR^a

Catalyst	Expected H ₂ Consumption Area (x10 ⁷ , a.u.)	Actual H ₂ Consumption Area (x10 ⁷ , a.u.)	Reduction Degree (%)
Fe-0Mn	1.680	1.680	100
Fe-2Mn	1.647	1.646	99.9
Fe-13Mn	1.464	1.306	89.2
Fe-31Mn	1.163	0.545	46.9

^a Assume that Fe-0Mn composed of Fe₃O₄ can be fully reduced to metallic Fe during the TPR, and no reduction of divalent Mn is involved for all the catalysts.

Table S2⁵⁷Fe Mössbauer parameters of the spent Fe-Mn microspheres after FTO reaction^a.

Spent catalysts	Phases	IS/mm s ⁻¹ ^b	QS/mm s ⁻¹ ^c	Hhf/ kOe ^d	Area % ^e
Fe-0Mn	Fe ₃ O ₄ (A)	0.28	0.00	497	10.8
	Fe ₃ O ₄ (B)	0.67	-0.01	464	21.5
	χ-Fe ₅ C ₂ (I)	0.24	0.12	216	27.1
	χ-Fe ₅ C ₂ (II)	0.20	0.07	109	16.2
	ε-Fe ₂ C	0.17	0.07	187	24.4
Fe-13Mn	Fe ₃ O ₄ (B)	0.42	0.07	464	16.4
	Fe ³⁺ (spm)	0.28	1.03	-	9.5
	χ-Fe ₅ C ₂ (I)	0.24	0.12	217	24.3
	χ-Fe ₅ C ₂ (II)	0.22	0.03	111	16.9
	ε-Fe ₂ C	0.17	0.05	187	33.0
Fe-31Mn	Fe ³⁺ (spm)	0.29	0.72	-	22.5
	Fe ²⁺ (spm)	1.09	0.51	-	34.2
	χ-Fe ₅ C ₂ (I)	0.26	0.12	213	12.1
	ε-Fe ₂ C	0.23	0.06	176	31.2

^a Reaction conditions: H₂/CO=1.0, 533 K, 2.0 MPa and 4000 h⁻¹; ^b isomer shift; ^c quadruple splitting; ^d Hyperfine magnetic field; ^e relative absorption area.

Table S3

Phase composition of the spent Fe-Mn microspheres from Mössbauer spectroscopy.

Spent catalysts	Total oxides, %	Total carbides, %	ϵ -Fe ₂ C, %	χ -Fe ₅ C ₂ , %	ϵ -Fe ₂ C/ χ -Fe ₅ C ₂ ratio
Fe-0Mn	32.3	67.7	24.4	43.3	0.56
Fe-13Mn	25.8	74.2	33.0	41.2	0.80
Fe-31Mn	56.7	43.3	31.2	12.1	2.58

Table S4FTO performance of Fe-Mn microspheres with different Mn contents^a.

Catalysts	CO	CO ₂	Hydrocarbon distribution, carbon based, mol % ^b										FTY ^c ,	C ₂₋₄ = STY ^d ,	O/P ^e
	conv., %	sel., %	CH ₄	C ₂ ⁼	C ₃ ⁼	C ₄ ⁼	C ₂₋₄ ⁼	C ₂ ⁰	C ₃ ⁰	C ₄ ⁰	C ₂₋₄ ⁰	C ₅ ⁺	μmol _{CO} ·g _{Fe} ⁻¹ ·s ⁻¹	g·kg _{cat} ⁻¹ ·h ⁻¹	ratio
Fe-0Mn	6.2	16.7	20.7	6.5	18.7	7.8	33.0	11.3	5.2	11.0	27.6	18.7	14.3	42.9	1.2
Fe-2Mn	5.6	19.2	18.6	9.2	22.6	10.5	42.4	10.9	5.8	9.7	26.4	12.6	12.8	48.5	1.6
Fe-13Mn	7.1	25.2	17.0	13.6	25.2	15.6	54.4	6.0	4.1	5.1	15.2	13.3	16.9	73.0	3.6
Fe-31Mn	3.3	17.8	22.4	12.7	24.0	13.1	49.8	6.4	4.2	6.2	16.8	11.0	10.9	34.2	3.0

^a Reaction conditions: 25mg catalyst diluted with SiC (1:3), 533K, 2.0 MPa, H₂/CO=1, GHSV=4000h⁻¹, TOS= 20h. ^b The hydrocarbon distribution is normalized on a carbon basis with the exclusion of CO₂. ^c Iron time yield (FTY) = mol of CO converted to hydrocarbons (excluding CO₂) per time (s) per weight of iron (g). ^d STY = space time yield. ^e olefin/paraffin ratio of C₂₋₄ hydrocarbons.

Table S5FTO performance at different H₂/CO ratios in syngas over Fe-13Mn catalyst^a.

H ₂ /CO ratios	CO conv., %	CO ₂ sel., %	Hydrocarbon distribution, carbon based, mol % ^b										FTY ^c , mmol _{CO} ·g _{Fe} ⁻¹ ·s ⁻¹	C ₂₋₄ = STY ^d , g·kg _{cat} ⁻¹ ·h ⁻¹	O/P ^e ratio
			CH ₄	C ₂ ⁼	C ₃ ⁼	C ₄ ⁼	C ₂₋₄ ⁼	C ₂ ⁰	C ₃ ⁰	C ₄ ⁰	C ₂₋₄ ⁰	C ₅₊			
0.5	4.2	32.7	13.2	16.0	26.9	17.3	60.2	4.4	3.1	4.0	11.5	15.1	12.2	57.7	5.2
1.0	7.1	25.2	17.0	13.6	25.2	15.6	54.4	6.0	4.1	5.1	15.2	13.3	16.9	73.0	3.6
3.0	11.8	17.9	29.6	10.3	19.6	10.4	40.3	9.3	6.8	5.5	21.5	8.6	13.7	43.8	1.9

^a Reaction conditions: 25mg catalyst diluted with SiC (1:3), 533K, 2.0 MPa, GHSV=4000 h⁻¹, TOS= 20h. ^b The hydrocarbon distribution is normalized on a carbon basis with the exclusion of CO₂. ^c Iron time yield (FTY) = mol of CO converted to hydrocarbons (excluding CO₂) per time (s) per weight of iron (g). ^d STY = space time yield. ^e olefin/paraffin ratio of C₂₋₄ hydrocarbons.

Table S6

FTO performance during a temperature-scanning experiment at temperatures ranging from 493-613 K over Fe-13Mn catalyst^a.

Temp., K	CO conv., %	CO ₂ sel., %	Hydrocarbon distribution, carbon based, mol % ^b										FTY ^c , mol _{CO} ·g _{Fe} ⁻¹ ·s ⁻¹	C ₂₋₄ = STY ^d , g·kg _{cat} ⁻¹ ·h ⁻¹	O/P ^e ratio
			CH ₄	C ₂ ⁼	C ₃ ⁼	C ₄ ⁼	C ₂₋₄ ⁼	C ₂ ⁰	C ₃ ⁰	C ₄ ⁰	C ₂₋₄ ⁰	C ₅₊			
493	6.5	10.3	11.3	6.5	11.7	7.8	26.0	3.5	2.3	2.0	7.7	54.9	19.7	40.2	3.4
533	10.6	24.1	19.7	11.0	20.2	10.9	42.0	6.5	4.0	2.7	13.2	25.1	27.4	91.1	3.2
573	20.2	37.6	28.7	9.3	21.1	10.0	40.4	9.4	4.5	2.7	16.6	14.3	42.8	137.2	2.4
613	42.1	47.6	33.6	5.4	21.1	8.3	34.8	15.5	6.5	2.8	24.7	6.9	75.1	207.0	1.4

^a Reaction conditions: 25mg catalyst diluted with SiC (1:3), H₂/CO = 1.0, 2.0 MPa, GHSV=4000 h⁻¹, holding time at each temperature: 5h. ^b The hydrocarbon distribution is normalized on a carbon basis with the exclusion of CO₂. ^c Iron time yield (FTY) = mol of CO converted to hydrocarbons (excluding CO₂) per time (s) per weight of iron (g). ^d STY = space time yield. ^e olefin/paraffin ratio of C₂₋₄ hydrocarbons.

Fig. S1.

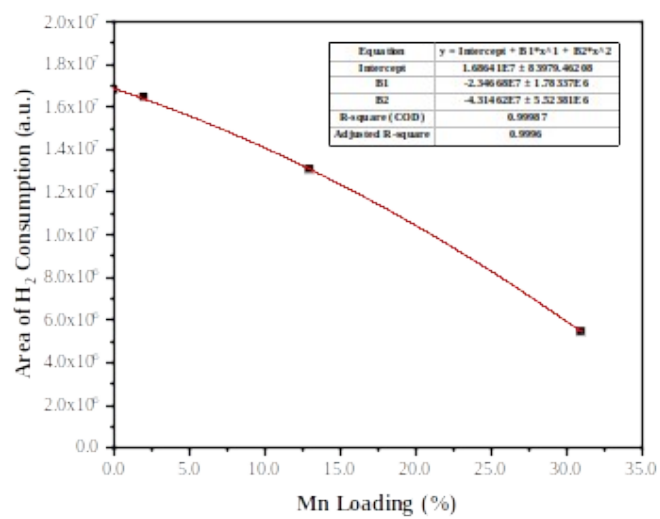


Fig. S1. Total area of H₂ consumption as a function of Mn loading from TPR experiments

Fig. S2.

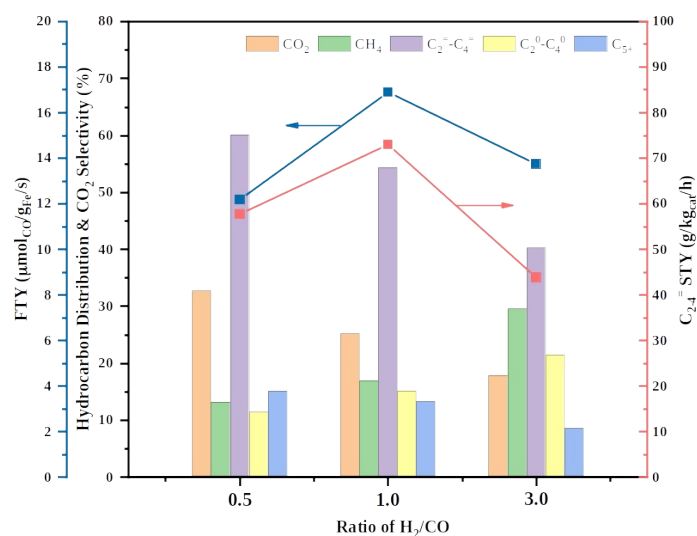


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Fig. S3.

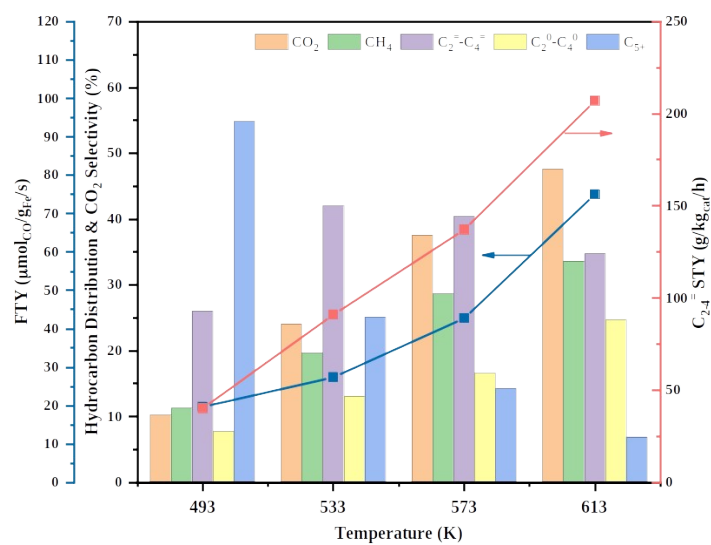


Fig. S3. Iron-time-yield (FTY), product distribution and C₂=-C₄= olefin space-time-yield (STY) during a temperature-scanning experiment in the range 493-613K over a Fe-13Mn catalyst. Reaction conditions: 25mg catalyst diluted with SiC (1:3), 2.0 MPa, GHSV=4000 h⁻¹, TOS= 20h, H₂/CO=1.

Mössbauer spectroscopy is a powerful technique for the identification and quantification of iron-based materials, especially for distinguishing poorly crystallized, microcrystalline, or amorphous phases [S1, S2]. From the spectra (Fig.9, Table S1), the spent Fe-0Mn could be fitted to two outermost sextets corresponding to Fe_3O_4 and three inner sextets, $\chi\text{-Fe}_5\text{C}_2$ (I), $\epsilon\text{-Fe}_2\text{C}$ and $\chi\text{-Fe}_5\text{C}_2$ (II), respectively. The two outermost sextets with isomer shift (IS) of 0.28, 0.67 mm/s were assigned to high-spin Fe^{3+} on tetrahedral sites (A-site) and a mixture of Fe^{2+} and Fe^{3+} on octahedral sites (B-site), denoted as Fe_3O_4 (A) and Fe_3O_4 (B) [S3]. The fitted sextets with the hyperfine field (Hhf) of 213-217 and 109-111 kOe are attributed to $\chi\text{-Fe}_5\text{C}_2$ [S4, S5], and the IS of 0.17-0.23 mm/s, Hhf of 176-187 kOe imply the presence of $\epsilon\text{-Fe}_2\text{C}$ [S5, S6]. For Fe-13Mn, only the Fe_3O_4 (B) (IS = 0.42 mm/s) exists, indicating that Mn^{2+} may preferentially substitute the tetrahedral sites in Mn-doped ferrite. Besides, the quadrupole doublet with IS of 0.28 mm/s and quadruple splitting (QS) of 1.03 mm/s is related to the superparamagnetic (spm) Fe^{3+} [S7]. Particularly, Fe-31Mn consists of two outer sextets assigned to $\chi\text{-Fe}_5\text{C}_2$, $\epsilon\text{-Fe}_2\text{C}$ and two inner doublets assigned to $\text{Fe}^{3+}(\text{spm})$, $\text{Fe}^{2+}(\text{spm})$. Large particles of Fe_3O_4 are wholly disappeared, carbide content drops sharply to 43.3% and the spm $\text{Fe}^{2+}/\text{Fe}^{3+}$ with small crystalline sizes prevail among iron species, probably due to the distortion and carburization restraining effect induced by Mn [S7].

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